Visual Analytics Project Proposal

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1 General idea

We are proposing to develop a visualization system with the purpose of providing general insights on the topic of vehicle accidents in Italy.

Intended users

The intended users for this system would be members of the Ministry of Transport, helping them with taking informed decision on how to better prevent accidents in Italy.

2 Dataset

The dataset in use contains details about vehicle accidents with injuries occurred in Italy during 2024. This dataset was extracted from the data published on the ISTAT website and comprises about 92.000 tuples for 8 attributes.

The attributes in the dataset are in summary: region and time (hour, weekday, month) of accident, road section where the accident occurred (traffic circle, intersection, etc...), type of accident (between vehicles, vehicle-pedestrian, etc...), deadliness of the accident, and number of accidents registered with the given characteristics.

More details are provided with a table at the end of the proposal.

3 Visualization System Description

The system is composed of two main sections as shown in Figure 1.

In section A it is visualized the number of accidents for each main attribute of the dataset. This is is done with a number of bar charts that use the categorical values on the bottom axis and show the number of accidents on the vertical axis. These charts are coordinated beetween each other. As an example, selecting the accidents that occurred between January and March also highlights the road section type and the kind of accidents occurred in the given timeframe.

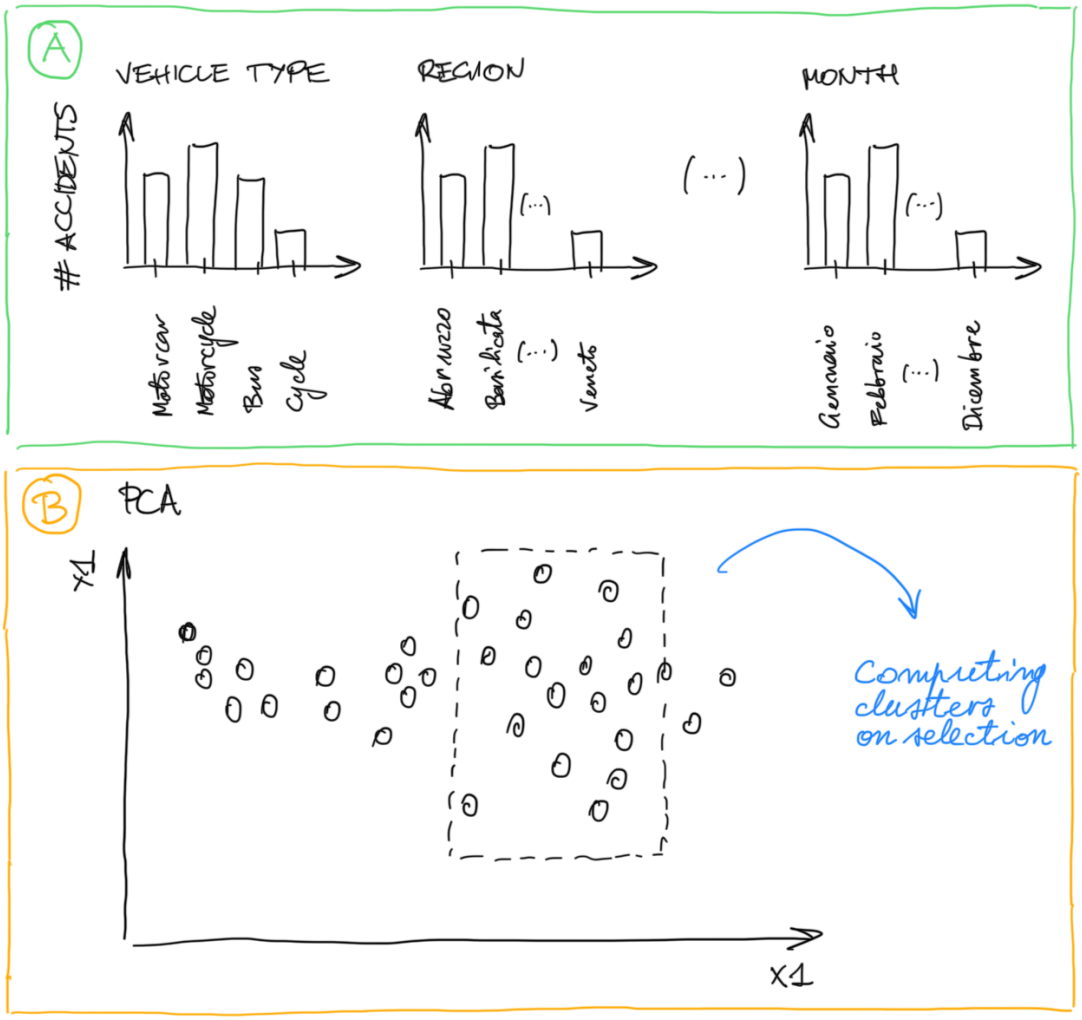

Figure 1: Draft of the visualization system.

In section B, it is presented to the user the result of PCA on the dataset. Data points can be colored according to the categorical attributes of the dataset: accident type, deadliness etc. The PCA graph is coordinated with the graphs in section A. As an example, selecting the accidents between January and March on section A higlights the relative data points on the PCA graph, and vice versa. Furthermore, the user can visually trigger an analytic by requesting the system to find clusters (*e.g. k-means*) in a selection over the PCA graph.

4 Dataset Details

Attribute	Description
Area	ID of Region
Road section	Number identifying a road section. Identifies: tunnel, bend, crossroad, traffic circle, bump/slope or bottleneck, straight stretch, level crossing.
Accident type	Number identifying the accident type. Identifies: accidents between vehicles, vehicle-pedestrian accients and accidents involving a single vehicle.
Deadliness	Boolean attribute that identifies if the accident involved death or only injury
Hour of accident	Time of accident measured with hour resolution
Month	Month number
Weekday	Number identifying the weekdays: monday, tuesday, wednesday, etc...
Observations	Number of accidents occurred with the specified attributes